Lingzhi Zhao

Ph.D. Candidate in Computer Science | University of Illinois Urbana-Champaign
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RESEARCH INTERESTS

Multimodal ML researcher with expertise in multimedia systems, focusing on **post-training of VLM and generative models** for **video applications**, i.e., video understanding, compression, streaming and quality assessment.

EDUCATION

University of Illinois Urbana-Champaign · Ph.D. in Computer Science — Aug. 2022 – May 2027
Advisor: Prof. Klara Nahrstedt

Shanghai Jiao Tong University · M.S. in Electrical and Computer Engineering — 2019 – 2022

Shanghai University · B.S. in Electrical and Computer Engineering — 2015 – 2019

TECHNICAL SKILLS

ML Frameworks	PyTorch, Transformers
Training & Inference	VLM/MLLM, SFT, RL post-training, on-device inference optimization
Programming & Tools	Python, C++, Java, Swift, Rust, Git, FFmpeg, NVCodec, LLMFarm

RESEARCH & WORK EXPERIENCE

TikTok, San Jose, CA — May 2026 – Present
Machine Learning Engineer Intern · Manager: Prof. Feng Qian

MLLM Post-training for Short-Video Quality Assessment *[Submitted to ICLR 2027]*

- Built a large-scale **short-video transcoding quality benchmark** with 377K subjective ratings which spans different source resolutions, transcoding settings, and mobile devices; evaluated SOTA MLLM scorers and systematically analyzed their failure.
- Post-trained a **context-aware MLLM via SFT with motion-artifact fusion**, achieving SOTA video quality prediction on the new benchmark and three public datasets.

Dynamic Ad Object Insertion for UGC Videos *[Manuscript in preparation]*

- Fine-tuned a Qwen3-VL embedding model via SFT for **brand-conflict detection and contextual similarity matching.**
- Accelerated SAM3 surface detection by **8.5**× through **adaptive frame sampling**.
- Built an **end-to-end automated ad-insertion** pipeline with **latency comparable to video transcoding**, serving as a proof of concept for online deployment.

University of Illinois Urbana-Champaign, MONET Lab — Aug. 2022 – Present
Graduate Research Assistant · Advisor: Prof. Klara Nahrstedt

AquaVLM: Mobile VLMs for Underwater Communication *[ACM UbiComp 2027] [Website]*

- Constructed an **85K-sample multimodal dataset** using GPT-4o few-shot conversation generation and fine-tuned **MobileVLM-V2 (3B) via SFT** to generate context-aware messages for divers underwater.
- Designed a **hierarchical message-generation** pipeline, achieving **>80% user-intent alignment** while reducing inference latency by **60%–70%**.
- Deployed the VLM on **iOS mobile devices** using 4-bit quantization with LLMFarm for an **interactive underwater communication prototype**.

AquaScope: Generative Compression for Mobile Image Transmission *[ACM MobiSys 2026] [Website]*

- Optimized **visual tokenizers (TiTok, VQGAN**) through **token clustering and fine-tuning** for **low-bandwidth, error-resilient** image transmission.

- Integrated the learned codec into a mobile acoustic prototype, improving **bandwidth efficiency** by **40%** and reducing average **BER** from **19% to below 2%**; outperformed PNG, JPEG, and classic neural compression.

GenStream: Generative Video Compression and Token Adaptation

- Optimized a **tokenizer-based generative** compression model for real-time video streaming, achieving **50%–78% bandwidth savings** over H.264/H.265 and prior neural codecs at comparable quality.
- Designed **content-aware token adaptation** for changing network conditions and improved perceptual quality (LPIPS) by **38%–59%** at matched bitrates and loss patterns versus state-of-the-art streaming systems.

ByteDance Inc., San Diego, CA May 2024 – Aug. 2025

Research Intern, Multimedia Lab · Manager: Dr. Shu Shi

MARS: Motion-Aware ABR for Real-Time VR *[Submitted to ACM MMSys 2027]*

- Developed a **perceptually guided bitrate adaptation** algorithm for a real-time VR streaming system, achieving **15% bandwidth reduction** without degrading QoE, supported by a user study.

Trinity: Low-latency Cloud VR Gaming *[ACM MMSys 2026] [Website]*

- Designed **action-aware rendering** based on measured latency tolerance, reducing bandwidth cost by **60%** and GPU utilization by **40%** while maintaining **QoE comparable to local streaming**.

SELECTED PUBLICATIONS

[1] **Lingzhi Zhao**, Yu Liu, Yilin Li, Diane Xu, Shi Han Chong, Yang Fan Mia, Xinyi Liu, Yufei Li, Xinheng Sheng, Isaac Zhang, Chen Zheng, Zhengyuan Yang, Lei Shi, and Feng Qian. Short UGC Video Quality Assessment via Large Language Model: Benchmark and Method. ***Submitted to ICLR 2027.***

[2] **Lingzhi Zhao**, Yongqiang Gui, Yanyan Suo, Sandesh Dhawaskar Sathyanarayana, Mingyuan Wu, Bo Chen, Ruixiao Zhang, Klara Nahrstedt, and Shu Shi. MARS: Streaming How You Move with Motion-Aware ABR for Real-Time VR. ***Submitted to ACM MMSys 2027.***

[3] Beitong Tian*, **Lingzhi Zhao***, Bo Chen, Haozhen Zheng, Jingcheng Yang, Mingyuan Wu, Deepak Vasisht, and Klara Nahrstedt. AquaVLM: Improving Underwater Situation Awareness with Mobile Vision Language Models. (*: equal contributions) ***ACM UbiComp 2027.*** *(corresponding author)*

[4] Beitong Tian*, **Lingzhi Zhao***, Bo Chen, Mingyuan Wu, Haozhen Zheng, Deepak Vasisht, Francis Y. Yan, and Klara Nahrstedt. AquaScope: Reliable Underwater Image Transmission on Mobile Devices. (*: equal contributions) ***ACM MobiSys 2026.*** *(corresponding author)*

[5] **Lingzhi Zhao**, Yongqiang Gui, Yanyan Suo, Sandesh Dhawaskar Sathyanarayana, Ruixiao Zhang, Shu Shi, and Klara Nahrstedt. Trinity: Exploiting Latency Sensitivity to Improve Quality of Experience on Cloud VR Gaming. ***ACM MMSys 2026.***

[6] **Lingzhi Zhao**, Ying Cui, Yuhang Jia, Yunfei Zhang, and Klara Nahrstedt. Enhancing Neural Adaptive Wireless Video Streaming via Lower-Layer Information Exposure and Online Tuning. ***IEEE Transactions on Multimedia 2025.***

[7] **Lingzhi Zhao**, Qian Zhou, Bo Chen, and Klara Nahrstedt. 360LiveCast: A Low-Latency and Bandwidth-efficient Multicast Framework for Live 360 Video. ***IEEE MIPR 2025.***

[8] **Lingzhi Zhao**, Ying Cui, Sheng Yang, and Shlomo Shamai (Shitz). An Optimization Framework for General Rate Splitting for General Multicast. ***IEEE Transactions on Wireless Communications 2022.***

[9] **Lingzhi Zhao**, Ying Cui, Zhi Liu, Yunfei Zhang, and Sheng Yang. Adaptive Streaming of 360 Videos with Perfect, Imperfect, and Unknown FoV Viewing Probabilities in Wireless Networks. ***IEEE Transactions on Image Processing 2021.***